

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Project Title	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Repository	Electricity-Consumption-Analysis
Website	Flask Web Application with Embedded Tableau Dashboard
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Project Description

The Electricity Consumption Analysis project analyzes electricity usage data collected from different states and regions of India. The project uses **MySQL** for database management, **Tableau** for interactive dashboard development, and **Flask** for deploying the dashboard through a web application.

The dashboard includes:

- State-wise Electricity Usage
- Monthly Electricity Usage Trend
- Region-wise Electricity Usage
- State-wise Electricity Usage Map
- Top 10 Electricity Consumers
- Business Insights & Conclusion
- Interactive Dashboard
- Tableau Story

The objective of this project is to help government organizations, electricity boards, researchers, and analysts understand electricity consumption patterns and support data-driven decision making.

Customer Persona

Primary User: Energy Analyst
Age: 25–50 Years
Occupation: Energy Analyst / Government Electricity Officer
Experience: Monitors electricity usage data, prepares reports, and supports energy planning.

Goals

- Analyze electricity consumption across states.
- Monitor monthly electricity usage.
- Compare electricity demand among regions.

- Identify high electricity consuming states.
 - Support better energy planning.
 - Make data-driven decisions using dashboards.
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Challenges

- Large electricity datasets are difficult to analyze manually.
 - Comparing multiple states requires significant effort.
 - Manual reports consume considerable time.
 - Traditional reports do not provide interactive visualization.
 - Identifying electricity consumption trends is difficult.
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Customer Journey Map (Five Stages)

Stage 1 — Access Dashboard

Experience

- Opens the Flask application.
- Loads the Tableau Dashboard.

Interactions

- Homepage opens successfully.
- Dashboard becomes available.

Goals

- Quickly access electricity data.

Positive Moment

- Dashboard loads quickly.

Negative Moment

- Unsure where to start.

Opportunity

- Add a welcome page explaining dashboard features.
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Stage 2 — Review Electricity Statistics

Experience

- Reviews electricity usage charts.

Interactions

- Examines state-wise usage.

Goals

- Understand electricity consumption.

Positive Moment

- Easy comparison between states.

Negative Moment

- Large datasets may confuse new users.

Opportunity

- Provide chart descriptions.
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Stage 3 — Analyze Trends

Experience

- Uses filters.

Interactions

- Selects regions and states.

Goals

- Identify electricity usage trends.

Positive Moment

- Interactive charts update instantly.

Negative Moment

- Multiple filters may overwhelm beginners.

Opportunity

- Add filter reset button.
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Stage 4 — Review Business Insights

Experience

- Reviews Tableau Story.

Interactions

- Reads conclusions.

Goals

- Understand electricity demand.

Positive Moment

- Story summarizes findings clearly.

Negative Moment

- Story requires updated data.

Opportunity

- Refresh dashboards regularly.
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Stage 5 — Decision Making

Experience

- Uses insights for planning.

Interactions

- Shares reports.

Goals

- Improve electricity management.

Positive Moment

- Data supports better planning.

Negative Moment

- Future demand prediction is unavailable.

Opportunity

- Integrate AI-based electricity forecasting.
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